MATHEMATICS (BCA-103)

SETS

Practice Set – 1

Ques 1. Represent the given sets in Roster Form.

(a) A = {x : x is an integer, –1 x < 5}.
(b) B = {x : x is an integer and – 3 ≤ x < 7}.
(c) C = {x : x is a Natural Number which is less than 6}
(d) D = {x | x is a positive integer which is less than 10 and 2^x – 1 is an odd number}

Ques 2. Write each of the following in set builder form.

(a) A = {5, 10, 15, 20}

(b) B = {l, 2, 3, 6, 9, 18}

(c) C = {P, R, I, N, C, A, L}

(d) D = {0}

(e) E = { }

Ques 3. Write down all the subsets of

(a) {8}
(b) {p, q}
(c) {1, 3, 5}
(d) ϕ
(e) {3, π, 0}

Ques 4. Find x and y, if (2x, x+y) = (6, 2).

Ques 5. Given set A = {a, b, c}, B = {p, q, r} and C = {x, y, z, m, n, t} which of following are considered as universal set for all the three sets.

(a) P = {a, b, c, p, q, x, y, m, t}

(b) Q = {ϕ}

(c) R = {a, c, q, r, b, p, t, z, m, n, x, y}

(d) S = {b, c, q, r, n, t, p, q, x, m, y, z, f, g}

Ques 6. Which of the following are examples of the null set

(a) Set of odd natural numbers divisible by 2

(b) Set of even prime numbers

(c) { x : x is a natural numbers, x < 5 and x > 7 }

(d) { y : y is a point common to any two parallel lines}

Ques 7. State, whether the given set is infinite or finite:

(a) {3, 5, 7, ….}

(b) {1, 2, 3, 4}

(c) {….., -3, -2, -1, 0, 1, 2}

(d) {20, 30, 40, 50, ………., 200}

(e) {7, 14, 21, …………, 2401}

(f) {0}

(g) (∅)

(h) {x | x is an even natural number less than 10, 000}

(i) {All people in the world}

(j) {x | x is a prime number}

(k) {x | x $\in$ N and x > 10}

(l) {….., -2, -1, 0}

Ques 8. If A = {3, 5, 7, 9, 10}, B = {7, 9, 10, 13}, and C = {10, 13, 15}. Find (A ∩ B) ∩ (B ∪ C).

Ques 9. Given, X = {a, b, c, d} and Y = {f, b, d, g}, determine the final value of X – Y and Y – X.

Ques 10. Given sets are A = {a, b}, and B = {a, b, c}. Is A ⊂ B? Find A ∪ B.

Ques 11. If U = {x : x $\in$ N, x ≤ 9}, A = {x : x is an even number, 0 < x < 10}, and B = {2, 3, 5, 7}, what will be the Set (A U B)?

Ques 12. If U = {a, e, i, o, u} and

- A = {a, e, i}
- B = {e, o, u}
- C = {a, i, u}

Verify whether the value is true or not: A ∩ (B - C) = (A ∩ B) - (A ∩ C).

Ques 13. Let U = {1, 2, 3, 4, 5, 6}, A = {2, 3} and B = {3, 4, 5}.

Find A′, B′, A′ ∩ B′, A ∪ B and hence show that (A ∪ B)′ = A′∩ B′.

Ques 14. Let U = {1, 2, 3, 4, 5, 6, 7}, A = {2, 4, 6}, B = {3, 5} and C = {1, 2, 4, 7}, find

(a) A′ ∪ (B ∩ C′)

(b) (B – A) ∪ (A – C)

Ques 15. Show that for any sets A and B,

(a) A = (A ∩ B) ∪ (A – B),

(b) $A \cup (B - A) = (A \cup B)$

Ques 16. Find the power set of the following sets.

(a) $A = \{a, b, c\}$

(b) $B = \{0, 7\}$

(c) $C = \{0, 5, 10\}$

(d) $D = \{x\}$

Ques 17. To prove that

(a) $(A \cup B)' = A' \cap B'$.

(b) $(A \cap B)' = A' \cup B'$.

Ques 18. If $A = \{1, 3, 5\}$ and $B = \{2, 3\}$, then find:

(a) $A \times B$

(b) $B \times A$

(c) $A \times A$

(d) $(B \times B)$

Ques 19. Out of 50 students of class IX, 25 students liked to play football, 35 liked to play basketball and 15 liked to play both the games. How many students do not like to play any games?

Ques 20. In a class of 25 students, 12 students have taken Economics, 8 have taken Economics but not Maths. Find (i) the numbers of students who taken Economics and Maths, (ii) those who have taken Maths but not Economics.